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Inertial Impaction Technique for the Classification of Particulate Matters and Nanoparticles: A Review †

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Abstract

Inertial impactors are applied widely to classify particulate matters (PMs) and nanoparticles (NPs) with desired aerodynamic diameters for further analyses due to their sharp cutoff characteristics, simple design, easy operation, and high collection ability. A few hundred papers have been published since the 1860s that addressed the characteristics and applications of the inertial impactors. In the last 30 years, our group has also carried out lots of studies to contribute to the design and the improvement of inertial impactors. With our understanding of inertial impactors, this article reviews previous studies of some typical types of the inertial impactors including conventional impactors, cascade impactors, and virtual impactors and the parameters for design consideration of these devices. The article also reviews some applications of the inertial impactors, which are mass concentration measurement, mass and number distribution measurement, personal exposure measurement, particulate matter control, and powder classification. The synthesized knowledge of the inertial impactor in this study can help researchers to design an inertial impactor with an accurate cutoff diameter, a sharp collection efficiency curve, and no particle bounce and particle overloading effects for long-term use for PM classification and control purposes.

Keywords: inertial impactor, cascade impactor, virtual impactor, nanoparticles, sampling

1. Introduction

There was a great demand to develop devices for collecting airborne particles of different sizes to study the effect of particulate matters (PMs) on human health (Anderson J.O. et al., 2012). In the period from 1860 to 1880, researchers discovered that inertial impactors can be used to collect airborne particles for further analysis (Marple V.A., 2004). The inertial impactors including the conventional impactor, cascade impactor, and virtual impactor can classify particles into different size ranges based on particle's inertia. In the inertial impactors, when the airflow is forced to change the direction, the particles with sufficient inertia escape the turning flow while the particles with less inertia remain in the flow resulting in the particle size-based classification. A burst of development of the inertial impactors has exploded since 1945 when the first cascade impactor (a series of conventional impactors) was developed to collect size-fractionated airborne particle samples with a high collection efficiency to obtain the particle mass distribution (May K.R., 1945). Since the 1970s, researchers computed the flow field and particle trajectories within the impactor by using the finite difference methods and showed that the impactor has a sharp cutoff characteristic (McFarland A.R. et al., 1978). Thereafter, the impactor was considered as a standard device for PM classification based on particle aerodynamic diameter.

The virtual impactor was developed later since 1966 (Conner W.D., 1966) with a feature not found in the conventional impactor, in which particles are collected in a probe instead of on the collection plate to avoid particle bounce. The virtual impactor also attracted lots of interest from researchers for PM classification since two particle size fractions can be collected for further analysis as the design theory was studied and better understood (Marple V.A. and Chien C.M., 1980; Zahir M.Z. et al., 2018). Since 1970 until now, inertial impactors have been used extensively for various purposes such as (i) size-based classification for mass concentration measurement (Le T.C. and Tsai C.J., 2017; Le T.C. et al., 2019), (ii) particle mass distribution measurement (Marple V.A. et al., 1991; Demokritou P. et al., 2004a), (iii) chemical composition measurement (Marple V.A. et al., 1991), (iv) personal exposure measurement (Tsai C.J. et al., 2012b), and (v) nanoparticle (NP) measurement (Chow J.C. and Watson J.G., 2007).

The main reasons behind the diverse application of the

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inertial impactors are their sharp separation, high collection abilities, relatively simple design, and easy operation. But several limitations are inherent to the impactors such as particle bounce and re-entrainment (Turner J.R. and Hering S.V., 1987; Fujitani Y. et al., 2006), particle overloading (Tsai C.J. and Cheng Y.H., 1995; Demokritou P. et al., 2004b), and particle loss (Fang C. et al., 1991; Durand T. et al., 2014), which can degrade their performance. Moreover, some typical characteristics of the inertial impactors such as the cutoff diameter, the sharpness of collection efficiency curve, and the properties of the impaction substrate should be understood well since PMs have different sizes, shapes, types, compositions and characteristics (Loo B.W., 1975; Hinds W.C., 1999). There have been many studies conducted to resolve these issues from 1970 onward, which are reviewed in detail in Section 2.

In the last 30 years, our group has also carried out lots of studies to contribute to the design and the improvement of inertial impactors. In 1995, an impactor with an inverted conical cavity as the collection plate was developed to eliminate the particle bounce and re-entrainment and enhance the loading capacity (Tsai C.J. and Cheng Y.H., 1995). This design was studied further by other researchers (Chang M. et al., 1999; Kim D.S. et al., 2006; Kim Y.J. and Yook S.J., 2011; Kim W.G. et al., 2013) and used for the development of the EPA (Environmental Protection Agency) $PM_{2.5}$ well impactor ninety-six (WINS) (Peters T.M. et al., 2001). Some different personal samplers were developed to measure respirable particles and NPs for the exposure assessment at workplaces such as personal inertial impactor (Huang C.H. et al., 2006) and novel active personal nanoparticle sampler (PENS) (Tsai C.J. et al., 2012b). The PENS was used for further studies by other research groups (Hsiao T.C. et al., 2010; Hsiao T.C. et al., 2011; Sung G. et al., 2018) or for field measurements (Young L.H. et al., 2013; Zhou Y. et al., 2014; Zhu C.S. et al., 2016; Asbach C. et al., 2017; Zhu C.S. et al., 2017). Some commercial sampling devices were studied to improve their sampling performance at our laboratory. We found the inter-stage loss of NP clogging in the last four stages of the micro-orifice uniform deposit impactor (MOUDI) due to its step-wised nozzle structure with abrupt contraction (Liu C.N. et al., 2013b) and developed the NCTU micro-orifice cascade impactor (NMCI) with smooth nozzle shape to eliminate these problems (Chien C.L. et al., 2015). The sampling performance of the standard EPA louvered PM_{10} inlet and $PM_{2.5}$ WINS inlet was studied and the particle bounce, re-entrainment, and particle overloading effects of these devices were found (Le T.C. and Tsai C.J., 2017; Le T.C. et al., 2019). The non-bouncing modified louvered PM_{10} (M-PM_{10}) inlet and the modified WINS (M-WINS) were developed for the long-term sampling purpose without the need for frequent maintenance. A multi-filter PM_{10} and $PM_{2.5}$ sampler (MFPPS) which consists of PM_{10} and $PM_{2.5}$ impactors was developed for collecting multiple samples of PM_{10} and $PM_{2.5}$ simultaneously for different analyses or comparison tests of different devices (Liu C.N. et al., 2011). Besides, numerous theoretical and experimental studies have been carried out by our group, which explored fundamental issues of the inertial devices including the effect of gravity on particle collection efficiency (Huang C.H. and Tsai C.J., 2001), the influence of impaction plate diameter and particle density on the collection efficiency of inertial impactors (Huang C.H. and Tsai C.J., 2002a), the effect of porous metal (Huang C.H. et al., 2001; Huang C.H. and Tsai C.J., 2002b; Huang C.H. and Tsai C.J., 2003) and porous foam (Huang C.H. et al., 2005) as the impaction substrate on the cut-size and collection efficiency, and the influence of the relative humidity (RH) on NP concentration and particle mass distribution measurements by the MOUDI (Chen S.C. et al., 2011).

With our understanding of inertial impactors, this article aims to synthesize the existing knowledge of impactor design and discuss future perspectives in the applications of inertial techniques to classify PMs, NPs, and powder. The design principle, cutoff characteristics, and the advantages and disadvantages of each type of inertial impactors are reviewed. Some typical applications of inertial impactors including mass concentration measurement, mass and number distribution measurement, personal exposure measurement, PM control, and powder classification are discussed. Two types of inertial impactors including the conventional impactor and the virtual impactor are covered.

2. Principle and design consideration

The types and principles of the inertial impactors are presented in textbooks (Hinds W.C., 1999; Marple V.A. and Olson B.A., 2011) and are described briefly here to support the design and application considerations. The description and pros and cons of each type of inertial impactors are summarized in **Table 1**. The lists of their commercial devices can be found in Watson J.G. and Chow J.C. (2011).

2.1 Conventional impactor and cascade impactor

A conventional single-stage impactor consists of a single or multiple nozzles which are either round- or rectangular-shaped and an impaction plate below the nozzle as shown in **Fig. 1a**. When the particle-laden air passes through the nozzle(s) and makes a turn above the impaction plate, particles with sufficient inertia are not able to follow streamlines and impact on the collection plate while particles with less inertia will flow out of the

Table 1 Types of inertial impactors.

Brief description		Pros and cons	Typical devices	References
Single-stage, single-nozzle impactor	The impactor consists of a single nozzle and an impaction plate, used to collect particles smaller than a certain size.	***Pros:*** Simple configuration, easy operation, and low cost. ***Cons:*** Low flow rate, large cut-size, and possible particle bounce, particle overloading, and particle loss.	EPA $PM_{2.5}$ WINS; EPA louvered PM_{10} inlet; EPA flat-topped PM_{10} inlet	McFarland A.R. et al., 1984; Peters T.M. et al., 2001; Tolocka M.P. et al., 2001
Single-stage, multi-nozzle impactor	The impactor consists of a nozzle plate with multiple nozzles and an impaction plate, used to collect particles smaller than a certain size.	***Pros:*** Simple configuration, easy operation, a wide range of flow rates, and a smaller cut-size. ***Cons:*** Particle bounce, particle overloading, particle loss, and multiple jet interactions.	EPA Hivol PM_{10} impactor (1133 L/min); Harvard sharp-cut impactor (10 and 20 L/min)	McFarland A.R. et al., 1984; Turner W.A. et al., 2000
Cascade impactor	The impactor is a series of multi-nozzle impactors, used to measure particle mass distribution.	***Pros:*** It can measure particle mass distribution ranging from 10 nm to 10 µm. ***Cons:*** Possible overloading, particle re-entrainment, and interstage loss.	MOUDI; MOUDI-II; Andersen cascade impactor; NMCI	Andersen A.A., 1966; Marple V.A. et al., 1991; Chien C.L. et al., 2015
Low pressure impactor (LPI)	The cascade impactor operates at low pressure condition.	***Pros:*** It can measure particle mass distribution of NPs down to 10 nm. ***Cons:*** High pressure drop.	Berner-type low-pressure impactor	Berner A., 1972
Electrical low pressure impactor (ELPI)	The device consists of a charger and a cascade impactor coupled with electrometers, used to measure particle number concentrations electrically	***Pros:*** Automatic and nearly real-time measurement with a response time of less than 5 s. ***Cons:*** Possible particle bounce unless the oil-coated sintered-metal substrate is used; Resolution is not good for NPs.	Dekati@ ELPI	Keskinen J. et al., 1992; Järvinen A. et al., 2014
Virtual impactor	The impactor consists of a single nozzle and a single collection probe, used to collect two particle size fractions.	***Pros:*** No particle bounce and particle loading effect; Simultaneous measurement of two particle size ranges. ***Cons:*** Contamination of coarse particles by particles smaller than the cut-size; Particle loss; Use of two flow meters.	Dichotomous sampler	McFarland A.R. et al., 1978

impactor and are either collected by a filter for further analyses or measured directly by an aerosol instrument (Marple V.A. and Olson B.A., 2011). In order to classify particles into many size ranges, a series of single-stage impactors are used, which is called cascade impactor (Marple V.A. and Olson B.A., 2011). The successive stages are designed to provide progressively higher jet speeds and/or reduced pressures so that the average aerodynamic size of particles collected at each stage is progressively smaller. Many nozzles (multi-nozzle) are mainly used to increase the airflow rate. These nozzles are drilled on a nozzle plate which is placed on the top of the impaction plate with a given jet-to-plate distance.

2.1.1 Design theory

The design theory of the conventional impactor is based on the numerical modeling of the flow field and the equations of particle motion by Marple and co-workers (Marple V.A. and Rubow K.L., 1986). The performance of the inertial impactor is characterized by the collection

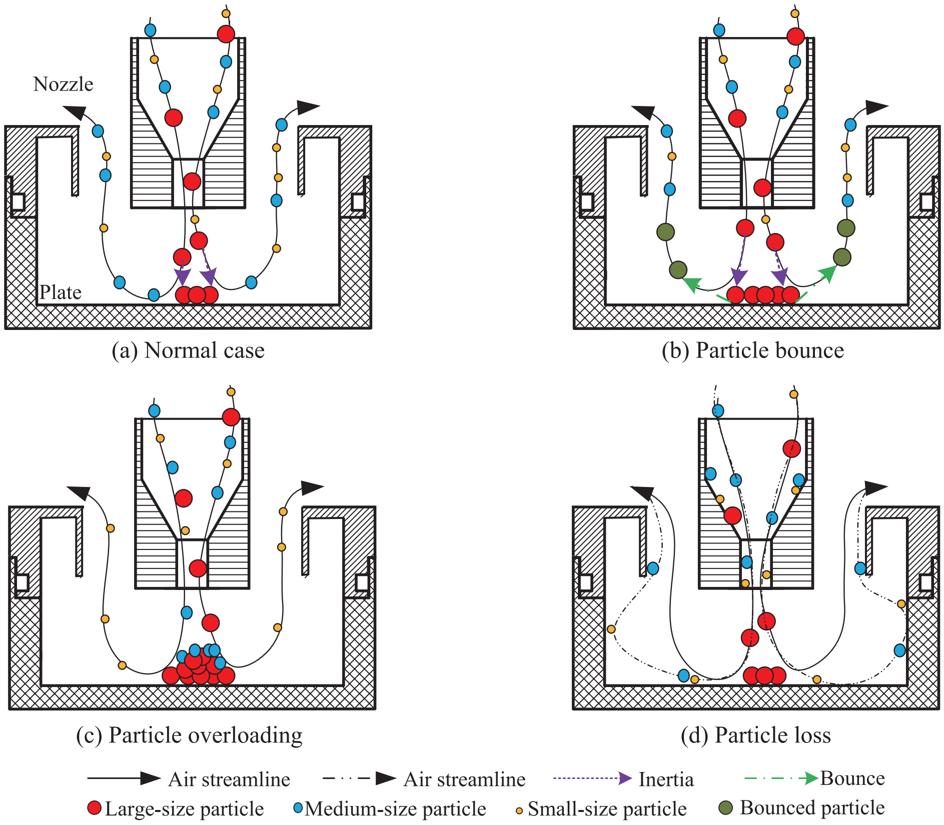

Fig. 1 Inertial impactor problems **(a)** normal case **(b)** particle bounce, **(c)** particle overloading, and **(d)** particle loss.

efficiency curve, which is an "S" shape curve due to the non-ideal and non-uniform flow field in the impactor. The collection efficiency curve is represented by the cutoff aerodynamic diameter (D_{pa50}), which is the aerodynamic diameter for 50 % collection efficiency, and the sharpness or geometric standard deviation (GSD ≥ 1.0) of the particle collection efficiency curve. The GSD is calculated as (Marple V.A. and Olson B.A., 2011)

$$\mathrm{GSD} = \sqrt{\frac{D_{pa84}}{D_{pa16}}} \tag{1}$$

where D_{pa84} and D_{pa16} are the aerodynamic diameters for 84 % and 16 % collection efficiencies, respectively. The most important parameter that governs the collection efficiency is the Stokes number (*Stk*) which is used to predict whether a particle will impact on the impaction plate or follow the air streamlines to exit the impactor. *Stk* is the ratio of the particle's stopping distance to the characteristic dimension of the nozzle and is calculated as follows (Marple V.A. and Olson B.A., 2011)

$$Stk = \frac{\rho_p D_p^2 C_c(D_p) U_0}{9\mu W} = \frac{\rho_{p0} D_{pa}^2 C_c(D_{pa}) U_0}{9\mu W} \tag{2}$$

where ρ_p is the particle density (kg/m^3), ρ_{p0} is the unit density (1000 kg/m^3), C_c is the slip correction factor, U_0 is the air jet velocity (m/s), D_p is the particle diameter (m), μ is the dynamic air viscosity (kg·s/m^2), and W is the diameter or the width of nozzle (m). Stk_{50} is the Stokes number for 50 % collection efficiency. For an impactor, Stk_{50} is usually fixed under optimized conditions, and D_{pa50} can be calculated as the following equation (Marple V.A. and Olson B.A., 2011):

$$D_{pa50} = \sqrt{\frac{9\mu W}{\rho_{p0} C_c(D_{pa50}) U_0}} \sqrt{Stk_{50}} \tag{3}$$

Jet Reynolds number (*Re*) is another parameter governing the flow field and the particle collection efficiency. *Re* influences the sharpness of the collection efficiency curve more than the cutoff diameter. *Re* of the round-nozzle impactor is expressed as (Marple V.A. and Olson B.A., 2011)

$$Re = \frac{\rho_a W U_0}{\mu} \tag{4}$$

where ρ_a is the air density (kg/m^3). The inertial impactors will have sharp and nearly the same collection efficiency curves with a fixed Stk_{50} of 0.24 (round nozzle) or 0.59 (rectangular nozzle) if *Re* ranges from 500 to 3000.

The *S*/*W* ratio, which is the jet-to-plate distance (*S*) divided by *W*, also influences the collection efficiency. An impactor will have a sharp efficiency curve if S/W is greater than 1.0 for a round-nozzle impactor and greater than 1.5 for a rectangular-nozzle impactor (Marple V.A. and Rubow K.L., 1986). S/W usually ranges from 1 to 5 to eliminate the effect on Stk_{50} for single nozzle impactors (Marple V.A. and Liu B.Y., 1974). Furthermore, in order to provide a sufficient time for the particles to accelerate to the fluid velocity, the throat length (T) divided to W (T/W) should be larger than 1.0 and the entrance of the nozzle should be tapered or conical (Marple V.A. and Willeke K., 1976). In addition, the effect of particle gravity on the increasing particle collection efficiency was found when Re is below 1500 (Huang C.H. and Tsai C.J., 2001). Other parameters influencing the collection efficiency curves are the impaction plate diameter, the particle density (Huang C.H. and Tsai C.J., 2002a) and the design of impaction surfaces (Tsai C.J. and Cheng Y.H., 1995; Tsai C.J. and Lin T.Y., 2000). A small impaction plate will cause an increase in the cutoff diameter (Huang C.H. and Tsai C.J., 2002a) while a small impaction well will decrease the cutoff diameter due to the secondary impaction on the wall of the well (Hu S. et al., 2007).

It is noted that the above design considerations are not only suitable for singe-nozzle, single-stage impactor but also applicable for the multi-nozzle, multi-stage impactor with nozzles far enough from each other. If the nozzles are too close, nozzle crossflow interaction occurs, which affects the collection efficiency of the multi-nozzle impactor (Fang C. et al., 1991; Kwon S.B. et al., 2002). The cross-flow parameter, which is a function of the number of nozzles, nozzle diameter, and nozzle cluster diameter, should be less than 1.2 to eliminate the effect of the cross-flow interaction (Fang C. et al., 1991). Other previous studies showed that the cutoff characteristics of the multi-nozzle impactor are also effected by S/W and Re even if they fall in the recommended range for the single-nozzle impactor (Gudmundsson A. et al., 1995; Kwon S.B. et al., 2002; Yao M. and Mainelis G., 2006). The cutoff characteristics of successive stages of the cascade impactor with the D_{pa50} < 100 nm was found to be very sensitive to S/W (Arffman A. et al., 2011; Tsai C.J. et al., 2012b; Liu C.N. et al., 2013b). In the multi-nozzle impactor, the secondary deposit of particles due to the jet interaction leads to the differences between the theoretical and experimented collection efficiencies and the less sharp collection efficiency and lower Stk (Rocklage J.M. et al., 2013; García-Ruiz E. et al., 2019).

To achieve smaller cut-sizes in the lower stages of the cascade impactor, multiple micro-orifice impactors (MOIs) or low pressure impactors (LPIs) are used. In the LPIs, particles with a very small cutoff diameter (e.g. 0.05 μm) are collected due to the increase of the slip correction factor with the decrease in the pressure. A large vacuum pump to draw the air through the nozzles at a high flow velocity is needed which creates a large pressure drop for the impaction to take place at low pressure (e.g. 3 kPa). The nano-size cutoff diameter (e.g. 2 or 5 nm) can also be achieved by expanding the aerosol into an evacuated region creating hypersonic impaction conditions (Fernandez de la Mora J. and Schmidt-Ott A., 1993). These low pressure impactors may have the evaporation loss of semi-volatile particle materials (SVM) (Biswas P. et al., 1987). In comparison, in the MOIs, the smaller cutoff diameter is obtained by using numerous small micro-orifices, which provide the desirable flow rate at a relatively low pressure drop. The evaporation loss of SVMs was shown to be eliminated in the MOIs if the pressure drop is less than 40 kPa (Fang C. et al., 1991). The cascade impactor can be developed with the above design considerations and calibrated with the standard method presented by Marple V.A. and Olson B.A. (2009). The consideration of designing an inertial impactor with a smaller cut-size, a sharper collection efficiency curve, and a higher flow rate is summarized in **Table 2**.

There major drawbacks of the conventional impactor are particle bounce, particle overloading and particle loss as shown in **Fig. 1b–d**. Since these effects can cause the shift of the cutoff diameter and the collection efficiency curve leading to incorrect PM measurement, numerous researches have studied these problems to find out the solutions (Liu C.N. et al., 2013b; Le T.C. and Tsai C.J., 2017; Le T.C. et al., 2019).

2.1.2 Particle bounce effect

Particle bounce occurs when the impactor plate is unable to absorb the kinetic energy of the incident particles completely (**Fig. 1b**). The particles are not retained by the surface, resulting in the lower particle collection efficiency for particles larger than the cutoff diameter (John W. et al., 1991; Bateman A.P. et al., 2013). The rebound particles can be collected on the after filter or a subsequent stage resulting in the positive bias of PM measurement or the shift of the mass distribution toward smaller aerodynamic sizes (Marple V.A. and Olson B.A., 2011; Jain S. and Petrucci G.A., 2015). The particle bounce depends on the adhesion energy between particles and the impaction surface, the energy loss mechanisms in particles (dissipation energy) and the initial velocity (or kinetic energy) of particles (Bateman A.P. et al., 2013; Jain S. and Petrucci G.A., 2015). After impaction, the incident kinetic energy is balanced by several energy relaxation terms including dissipation energy, adhesion energy, and rebound energy. Rebound energy that is greater than the sum of dissipation and adhesion energies will result in particle bounce. Physical properties of particles such as hygroscopicity, size, density, phase, elasticity, and hardness can also play an important role in particle bounce behavior (Matthew B.M. et al., 2008; Bateman A.P. et al., 2013;

Table 2 Design considerations of inertial impactors.

Requirements	Solutions	Limitations	References
To decrease the cut-size by	increasing the flow rate	*Re* numbers should be in the range of 500–3000 to achieve a sharp collection efficiency curve	Marple V.A. and Rubow K.L., 1986
	decreasing the diameter or width of nozzles	It may cause an increased pressure drop and it is hard to fabricate the micro-size nozzles	Berner A., 1972; Marple V.A. et al., 1991
	decreasing the jet-to-plate distance	It may cause an increased pressure drop if S/W is ≤ 1.0 for round-nozzle impactors and ≤ 1.5 for rectangular-nozzle impactors	Chien C.L. et al., 2015
	increasing nozzle throat length	$T/W \geq 1.0$	Marple V.A. and Liu B.Y., 1974
	increasing the impaction plate diameter	It is applicable as $Re < 1500$	Huang C.H. and Tsai C.J., 2002a
	using the elliptical concave impaction plate	It is applicable for round-nozzle impactor	Kim Y.J. and Yook S.J., 2011
	gravitation force assistance	It is applicable as $Re < 1500$	Huang C.H. and Tsai C.J., 2001
	reducing the pressure by using high jet velocities	Evaporation of semi-volatile compounds	Chen S.C. et al., 2007; Arffman A. et al., 2011
To increase the sharpness of the collection efficiency curve by	reducing the nozzle throat length	$T/W \leq 1.0$. It is just applicable for low pressure impactors	Arffman A. et al., 2011
	using an additional punched impaction plate	It is a complicated configuration	Cheon T.W. et al., 2017
	using horizontal inlet with rectangular-slit-nozzle	It is a complicated configuration	Kim M.K. et al., 2014
To increase the flow rate by	increasing the number of nozzles	W should be ≥ 0.04 cm to lower fabrication cost	Chien C.L. et al., 2015
	using rectangular nozzles	The less sharp collection efficiency curve was found	Kim M.K. et al., 2014

Kang M. et al., 2015; Chen M. et al., 2016). It was reported that liquid and liquid-coated particles were collected with an efficiency of nearly 100 %, while the collection efficiency of solid particles could be as low as 20 % (Matthew B.M. et al., 2008).

To increase the adhesion energy and minimize the solid particle bounce, the impaction plate is usually coated with vacuum grease, vacuum oil or silicone oil (Turner J.R. and Hering S.V., 1987), covered by a glass-fiber filter (GFF) saturated in water (Dunbar C. et al., 2005), or soaked in vacuum oil (Vanderpool R.W. et al., 2007). Use of rough impaction plates (Marjamäki M. and Keskinen J., 2004), porous substrate (Huang C.H. et al., 2001; Huang C.H. and Tsai C.J., 2003; Huang C.H. et al., 2005), or specially designed impaction plate (Kim Y.J. and Yook S.J., 2011; Kim W.G. et al., 2013) can also reduce particle bounce. For the coated substrate, the type and the thickness of the coating material also affect the particle bounce. If the coating materials interfere with chemical analysis (Wang L. et al., 2005; Fujitani Y. et al., 2006), an uncoated impaction surface is preferred (Kang M. et al., 2015). In addition, the relative humidity of incoming aerosols conditioned to a higher value (> 70 %) can minimize particle bounce, due to the increase in the adhesion energy by capillary force between adsorbed water on the particle and the impaction plate (Chen S.C. et al., 2011; Bateman A.P. et al., 2013).

2.1.3 Particle loading effect

Although the coating materials help to reduce the particle bounce effectively, the particle loading effect occurs when too many particles are collected on the impaction substrate after a long sampling period (**Fig. 1c**). When the grease-coated impaction plate is loaded heavily with particles, the incident particles strike on previously deposited solid particles rather than the coated plate, resulting in possible particle bounce (Tsai C.J. and Cheng Y.H., 1995). Moreover, the particle blow-off problem may take place when heavily loaded particles are swept away from the substrate by high-velocity jet flow.

Therefore, the net effects of substrate overloading are similar to particle bounce: lower collection efficiencies for particle larger than the cut-size and the shift of the measured size distribution toward the smaller particle sizes (Tsai C.J. et al., 2012b). The problem of particle overloading can be overcome by reducing the sampling time since it affects the amount of particles collected. Many methods have been proposed to increase the particle loading capacity on the impaction substrate to extend the sampling period, including the special designs of impaction plates (Tsai C.J. and Cheng Y.H., 1995), oil-soaked filters (Peters T.M. et al., 2001), oil-coated impaction substrate (Tsai C.J. et al., 2012b), multi-hole impactor with Trypticase Soy Agar (TSA) substrate (Lai C.Y. et al., 2010) or rotating substrates (Marple V.A. et al., 1991; Tsai C.J. et al., 2012b).

However, when the oil-coated filter substrate is used to eliminate particle bounce, the particle loading effect can still occur since the oil, which has a lower viscosity than the grease, wicks up through the previously deposited particle layer by the capillary action, and eventually leads to the pile-up of the deposited particles on the substrate. This causes the shift of the cutoff diameter and collection efficiency curve to the smaller sizes leading to incorrect PM measurements (Kenny L.C. et al., 2000; Vanderpool R.W. et al., 2001). Even with the rotating stage design in the MOUDI (Marple V.A. et al., 1991), it is expected that several tens of layers of deposited particles will appear on the grease-coated stage for particle bounce and loading effect to occur when the typical stage inlet aerosol mass concentration is 10–20 μg/m^3 during 24-h sampling (Tsai C.J. and Cheng Y.H., 1995). The special designs of impaction cavities and wells have better performance compared to that of flat impaction surfaces since rebound particles can be retained (Tsai C.J. and Cheng Y.H., 1995; Peters T.M. et al., 2001).

An M-PM$_{10}$ impactor with oil-soaked GFF substrate supported by a porous metal disc as shown in **Fig. 2** was designed to avoid the particle bounce and overloading effects (Le T.C. et al., 2019). In the M-PM$_{10}$ impactor, there is a 1-mm layer of oil remaining on the top of the substrate and the pores of the porous metal disc are filled with vacuum oil to enhance the loading capacity. With these improvements, the modified PM$_{10}$ impactor showed a good sampling performance with a bias of less than ± 10 % during 30 sampling days. Based on the concept of water-saturated GFF substrate of Dunbar C. et al. (2005) and impaction well (Peters T.M. et al., 2001), a PM$_{2.5}$ impactor with the water-wetted GFF substrate modified from the WINS was developed by our group as shown in

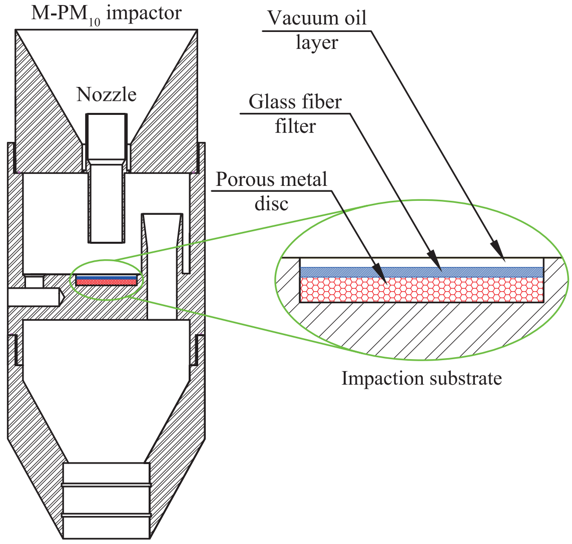

Fig. 2 Schematic diagram of the M-PM$_{10}$ impactor (Le T.C. et al., 2019).

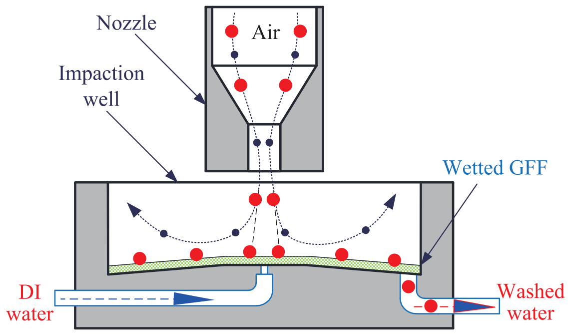

Fig. 3 Schematic diagram of the M-WINS (Le T.C. and Tsai C.J., 2017).

Fig. 3. This impactor design not only can eliminate the particle bounce by using the water-wetted GFF impaction substrate but also avoid the particle overloading effect by injecting a small amount of water upward to wash the deposited particles off the impaction surface continuously (Le T.C. and Tsai C.J., 2017). Instead of using water, the M-WINS can inject vacuum oil periodically to wash off particles to save oil consumption. Washing impaction substrate to eliminate particle bounce and particle overloading effects can be applied to different types of impactors as well.

2.1.4 Particle loss effect

Particle loss is another limitation of inertial impactors (**Fig. 1d**). Instead of being collected on the impaction substrate, particles can also deposit on the inner walls of the impactor, including the nozzle and the casing, leading to the underestimation of the collection efficiency and the distortion of the measured mass distribution. Large particles can be lost by impaction while nanoparticles can be lost by convention-diffusion. The losses may vary with particle properties. If particles are liquid or sticky, they will adhere to any surface upon contact with it. Some of dry particles, however, may rebound when they hit the surface and remain airborne. Experimental and theoretical studies have been conducted for determining the particle losses by impaction and diffusion (Chen S.C. et al., 2007). The inertial loss of large particles normally occurs on the contractions which connect the big inlet tube to the small nozzle or on the surface of nozzle plate (Durand T. et al., 2014; Kumar A. and Gupta T., 2015a, 2015b).

Another particle loss in the cascade impactor is the inter-stage particle loss, which occurs in the inter-stage space above the intended stage to collect particles. The inter-stage loss is due to turbulent particle deposition when they pass through sharp corners or obstructions between two adjacent stages (Marple V.A. et al., 1991). Typically, significant particle loss occurs in the lower stages of the cascade impactors due to diffusion-convention (Demokritou P. et al., 2002b; Watson J.G. and Chow J.C., 2011). The NMCI with smooth nozzles could minimize the convection-diffusion particle deposition in the nozzle (Liu C.N. et al., 2013b). In another study, 80 % of particle loss was found to concentrate around the nozzles and 20 % on the impaction plate in a stage of the DLPI (Dekati® Low-Pressure Impactor) (Durand T. et al., 2014).

Some conventional impactors were designed by changing the geometry of the impactor to enhance the collection efficiency to obtain a lower cut-size, including using the horizontal inlet to concentrate particles in the central axis of nozzle (Kim M.K. et al., 2014; Kim W.G. et al., 2014; Heo J.E. et al., 2018), using an additional punched impaction plate installed between the nozzle outlet and impaction plate (Cheon T.W. et al., 2017), and the combination of the impactor and the initial filter to achieve a sharp collection efficiency curve (Zhang T. et al., 2017; Zhang X. et al., 2018). **Table 3** lists the design solutions to resolve the problems of conventional impactors.

2.2 Virtual impactors

The virtual impactor is another type of inertial impactors, which consists of the nozzle(s), and the collection probe(s) (Marple V.A. and Olson B.A., 2011). In the virtual impactor, the collection probe diameter (D), which is slightly larger than W, is used to replace the collection plate of the conventional impactor to eliminate particle bounce and particle overloading effects (Loo B.W., 1975). Particles with sufficient inertia are retained in a relatively stagnant air passing through the collection probe while particles with less inertia follow the airflow to exit the impactor. The minor flow (Q_1) is about 5–20 % of the total air flow (Q) or the major flow (Q_o) is about 80–95 % of the total air flow (Hinds W.C., 1999).

Many studies have been conducted to examine the cutoff characteristics and the influencing parameters of the virtual impactor of the basic design (McFarland A.R. et al., 1978; Marple V.A. and Chien C.M., 1980; Lim K.S. and Lee K.W., 2006). They concluded that the theoretical cutoff characteristics of the virtual impactors are similar to those of conventional impactors, which are mainly governed by *Stk* and *Re* (Marple V.A. and Chien C.M., 1980). The other parameters such as collection probe/nozzle diameter ratio, nozzle length, entrance cone angle, and nozzle to collection probe distance do not have an important effect on the collection efficiency. Although the value of D/W does not affect the cutoff characteristic, it should be less than 1.49 to eliminate the flow field effect. The cutoff diameter increases as the minor and major flow ratio (Q_1/Q_o) decreases (McFarland A.R. et al., 1978; Marple V.A. and Chien C.M., 1980). The virtual impactor avoids the particle bounce effect without using the impaction plate, but an intrinsic disadvantage of the virtual impactor is the contaminations of coarse particles in the major flow by fine particles. That is, a portion of particles less than the cutoff diameter will remain in the minor flow or the so-called minor flow contamination. A virtual impactor with reduced fine particle contamination was designed by confining the aerosol flow within a central core enveloped by a sheath of clean air (Chein H. and Lundgren D.A., 1993). The opposing jets design and the introduction of a clean air core as the envelope around the particle-laden stream were also used to eliminate fine particle contamination.

Another major disadvantage of the virtual impactor is the particle internal losses which affect the cutoff diameter significantly (Xu Z., 1991). Particles are normally lost at the upper edge of the collection probe or on the backside of the probe plate. The particle loss is affected by D/W, Re, Q_1/Q, the nozzle protruding through the nozzle plate, the shape of the collection probe inlet, and the alignment of the axes of the nozzle and the collection probe. High particle losses occur for particles having a diameter close to the cutoff point (Marple V.A. and Chien C.M., 1980). The particle loss can be reduced by proper contouring of the probe entrance, decreasing Re, and increasing Q_1/Q_o and S/W ratio (Marple V.A. and Chien C.M., 1980). The particle loss can be reduced by increasing D/W but it may reduce the sharpness of the particle collection efficiency curve. Certain values of the throat length of the probe opening, the radius of the rounded shoulder of the probe opening, and the diameter of the lower surface of the nozzle required to reduce the particle loss were suggested (Marple V.A. and Chien C.M., 1980). It is summarized that D should be about 30–50 % larger than W, the radius on the inside of the collection probe inlet is about 30 % of W, the nozzle should protrude two to three times W through the nozzle plate, S should be about 1–1.8 times W, and the axes of the nozzle should be

Table 3 Solutions of inertial device problems.

Problems	Solutions	Pros and cons	References
Particle bounce	Using glass fiber filter or Teflon filter as an impaction substrate	***Pros:*** Simple and no chemical interference ***Cons:*** Expensive	Marple V.A. et al., 1991
	Greased substrate	***Pros:*** Easy implementer ***Cons:*** Can't be suitable for chemical analysis; Particle overloading effect	Turner J.R. and Hering S.V., 1987; John W. et al., 1991
	Oiled substrate	***Pros:*** Cheap, high loaded mass, low viscosity, and uniform coating ***Cons:*** Particle loading effect and oil evaporation	Turner J.R. and Hering S.V., 1987; Peters T.M. et al., 2001; Le T.C. et al., 2019
	Porous metal substrate	***Pros:*** No chemical interference ***Cons:*** The shift of the collection efficiency curve and cutoff diameter; Less sharp curve	Huang C.H. et al., 2001; Huang C.H. and Tsai C.J., 2002b; 2003
	Foam substrate	***Pros:*** No chemical interference ***Cons:*** The shift of the collection efficiency curve and cutoff diameter; Less sharp curve	Huang C.H. et al., 2005
	Water substrate and water washing	***Pros:*** No chemical interference; Can be used to collect samples for chemical analysis; Long-term use ***Cons:*** Pumps are needed to inject and drain water	Dunbar C. et al., 2005; Le T.C. and Tsai C.J., 2017
	Cavity-type impaction substrate	***Pros:*** No energy, solutions or materials are needed ***Cons:*** The curve is not very sharp	Tsai C.J. and Cheng Y.H., 1995
	Elliptical concave impaction plate	***Pros:*** No chemical interference ***Cons:*** The determination of dimensions is needed; It just helps trap some re-bound particles	Kim W.G. et al., 2013
	Cup impactor	***Pros:*** No chemical interference ***Cons:*** Special design; The cut-size can't be predicted	Kim D.S. et al., 2006
	Using virtual impactor	***Pros:*** Two size fractions ***Cons:*** Fine particle contamination; High particle loss near the cut-size.	McFarland A.R. et al., 1978; Marple V.A. and Chien C.M., 1980
Particle overloading	Oiled substrate	***Pros:*** Cheap and high loaded mass ***Cons:*** Short-term use and oil evaporation	Turner J.R. and Hering S.V., 1987
	Porous metal substrate	***Pros:*** High loaded mass ***Cons:*** The shift of the collection efficiency curve and cutoff diameter; Less sharp curve	Huang C.H. et al., 2001
	Continuous water washing	***Pros:*** Long-term use without frequent maintenance ***Cons:*** Pumps are needed to inject and drain water	Le T.C. and Tsai C.J., 2017
	Rotated impaction plate	***Pros:*** Long-term use ***Cons:*** A motor is needed; Coated substrate; Complicated configuration	Marple V.A. et al., 1991; Tsai C.J. et al., 2012b
Particle loss	Horizontal inlet in virtual impactor	***Pros:*** High collection efficiency ***Cons:*** Complicated configuration	Heo J.E. et al., 2018
	Smooth nozzles	***Pros:*** No particle clogging ***Cons:*** Need special technique to fabricate micro-size nozzles	Fang C. et al., 1991; Liu C.N. et al., 2013b; Chien C.L. et al., 2015
Multiple-jet interaction	The symmetrical arrangement of nozzles	***Pros:*** Uniform deposition ***Cons:*** Not enough study	Kwon H.B. et al., 2018

aligned closely to that of the collection probe (Hinds W.C., 1999).

Some special designs of the virtual impactor to eliminate particle loss and improve the collection efficiency were developed. This includes the slit virtual impactor assembled with an orifice installed upstream of the acceleration nozzle to concentrate the minor flow at the center of the collection probe (Lim K.S. and Lee K.W., 2006; Lee H. et al., 2014) or the impactor with three-partitioned horizontal inlet (Zahir M.Z. et al., 2018). The virtual impactor with an ultrafine cutoff diameter at a moderate pressure drop was developed by using a very small round nozzle and at a low flow rate (Sioutas C. et al., 1994b) or slit nozzles to operate at a high flow rate (Sioutas C. et al., 1994a).

Since it is hard to control both minor and major flows, the virtual impactor is usually designed as a single-stage virtual impactor. The multi-stage virtual impactor to collect particles in different size ranges simultaneously was studied (Marple V.A. et al., 2013, 2014a; Wada M. et al., 2017). To increase the flow rate to 1000 L/min, the use of multi-nozzle virtual impactors was also designed to reduce the pressure drop (Szymanski W.W. and Liu B.Y., 1989). It is noticed that the multi-nozzle virtual impactor may have multiple jet interaction problem leading to the degradation of the impactor's performance (Fang C. et al., 1991). The virtual impactor with the rectangular nozzle can increase the flow rate without changing the cut-size as compared to the single round-nozzle virtual impactor (Masuda H. and Nakasita S., 1988). A slit-shaped nozzle virtual impactor in which the width of the slit is the same with the diameter of the round nozzle was developed to overcome the crossflow interaction problems of the multiple jets at a high flow rate (Ding Y. and Koutrakis P., 2000).

3. Application of inertial impactors

The development of the inertial impactor is in line with the need to develop PM measurement devices. PM measurement serves various purpose such as (i) the determination of PM_{10} or $PM_{2.5}$ concentrations to determine the compliance with the national ambient air quality standard (NAAQS), (ii) the determination of particle mass distribution, (iii) the determination of physical and chemical properties of particles, and (iv) the exposure assessment of PMs. Many studies were carried out by our group for ambient mass measurement of $PM_{2.5}$ and ultrafine particles using commercial inertial impactors for $PM_{2.5}$ mass concentration measurement and the evaluation of the performance of $PM_{2.5}$ monitors (Liu C.N. et al., 2013a), the measurement of mass concentration and chemical composition of $PM_{2.5}$ (Liu C.N. et al., 2014; Liu C.N. et al., 2015) and ultrafine particles (Chen S.C. et al., 2010a; Chen S.C. et al., 2010b), long-term measurement for $PM_{2.5}$ and ultrafine particles mass concentrations (Lin G.Y. et al., 2015), and the measurement of PM_{10}, $PM_{2.5}$ and $PM_{0.1}$ for source characterization and apportionment (Gugamsetty B. et al., 2012).

3.1 PM mass measurement

In order to measure the ambient PM mass concentration to determine the compliance with NAAQS for PM_{10} and $PM_{2.5}$, respectively, the PM FRM (Federal Reference Method) samplers and FEM (Federal Equivalent Method) monitors are used (U.S.EPA, 2017a; Le T.C. et al., 2020). The FRM samplers and FEM monitors use size-selective inlets to classify the desired particle sizes for collecting on the filter for further analysis. Due to the sharp cutoff characteristics of the inertial impactors, they are widely deployed for the size-selective inlets. The PM_{10} and $PM_{2.5}$ size-selective inlets are designed based on the U.S. EPA requirements for the cutoff diameter, the sharpness of the collection efficiency curve, the flow rate, and the accuracy for long-term sampling. A list of PM_{10} and $PM_{2.5}$ size-selective inlets including the conventional impactor and virtual impactor is shown in Table 26.1 in Watson J.G. and Chow J.C. (2011) and is not repeated here. U.S. EPA designated that the PM_{10} size-selective inlets should have the cut-size of 10 ± 0.5 μm and the sharpness of the sampling efficiency curve of 1.5 ± 0.1 (U.S.EPA, 2017a) and the $PM_{2.5}$ size-selective inlets should have the cut-size of 2.5 ± 0.2 μm but the curve sharpness is not specified. The sharpness curve of the $PM_{2.5}$ inlet needs to agree with that of the WINS (GSD = 1.18) (Peters T.M. et al., 2001) and very sharp-cut cyclone (VSCC, GSD = 1.16) (Kenny L.C. et al., 2004), which are the standard inlets specified in U.S.EPA (2017a). Other countries specified the curve sharpness of 1.2 ± 0.1 (MEPC, 2013; ECS, 2014).

3.1.1 The inlet for a single size fraction

The conventional impactors are used extensively as an inlet to classify a single size fraction since they have sharper cutoff characteristics than that of the cyclone, which include AMI (Airmetrics) MiniVol™ impactors (AMI, 2020), Harvard sharp-cut impactors (Turner W.A. et al., 2000), Hi-vol PM_{10} samplers (McFarland A.R. et al., 1984), Med-vol (Medium volume) PM_{10} samplers (Olin J.G. and Bohn R.R., 1983), Flat Top Dichot PM_{10} inlet (McFarland A.R. et al., 1978), BGI FRM Louvered PM_{10} samplers (Tolocka M.P. et al., 2001), and EPA WINS $PM_{2.5}$ inlet (Peters T.M. et al., 2001). These inlets are different in the sampling flow rate, cutoff diameter, and sampling curve sharpness. The Hi-vol PM_{10} sampler is widely used for sample PM_{10} with a high flow rate of 1133 L/min (Fujitani Y. et al., 2017). The impaction plate of the Hi-vol

PM_{10} sampler was suggested to be grease-coated to eliminate the particle bounce and re-entrainment to avoid PM_{10} overestimation as compared to other samplers (Lee S.J. et al., 2005). A slight difference in the cutoff characteristic of the PM_{10} inlet can cause a sensible difference in PM_{10} measurements between different PM_{10} inlets since the mode of many ambient mass-size distributions is nearly 10 µm (Ranade M.B. et al., 1990). The Med-vol PM_{10} sampler is the one used for sampling PM_{10} at the medium flow rate of 113 L/min (Gertler A.W. et al., 1993) but is not deployed widely. Some other impactors have been designed to operate at high flow rates that exceed hundreds of liters per minute (Misra C. et al., 2002) and even thousands of liters per minute (Fulghum M.R. et al., 2012; Staymates M. et al., 2013) to collect more mass of PM_{10}, $PM_{2.5}$, PM_1 or $PM_{0.1}$.

Nowadays, many FRM samplers and FEM monitors use low-volume flat-topped (McFarland A.R. et al., 1984) and louvered PM_{10} inlets (Tolocka M.P. et al., 2001) to classify particles smaller than 10 µm at the low flow rate of 16.7 L/min (U.S.EPA, 2017b). However, several studies found that the low-volume PM_{10} inlet has uncertainty in the measured sampling efficiency. These studies found that the sampling performance of PM_{10} inlet is very sensitive to variation of particle sources (Wang L. et al., 2005), particle size distribution (Buser M.D. et al., 2008), and TSP/PM_{10} ratio (Total suspended particulates) (Watson J.G. et al., 2011). The low-volume PM_{10} sampler overestimated PM_{10} when it was used to measure fly ash particles as compared to the Andersen cascade impactor (Park J.M. et al., 2009). When the particle source is dominant by particles larger than the cut-size, a small fraction of particles larger than the cut-size may cause the overestimation (Faulkner W.B. et al., 2014). In the previous study, we found that the distortion of the sampling performance of the low-volume PM_{10} sampler could be due to particle bounce and re-entrainment from the uncoated surface of the PM_{10} inlet, and particle overloading of the PM_{10} impactor after a long sampling period, resulting in over-measurement and under-measurement, respectively. Thus, the PM_{10} inlet was required to be cleaned every day to eliminate these effects. Similarly, most of FRM samplers and FEM monitors (16.7 L/min) using $PM_{2.5}$ WINS as the $PM_{2.5}$ inlet (U.S.EPA, 2017b), whose impaction well have to be replaced after 3–5 days sampling to eliminate the particle overloading problems (Le T.C. and Tsai C.J., 2017). The M-PM_{10} inlet and $PM_{2.5}$ M-WINS inlet developed by our group were tested for one month without frequent cleaning and maintenance. They can be used to replace the traditional PM_{10} inlets and $PM_{2.5}$ WINS inlet for PM sampling.

The MiniVol impactor with the flow rate of 5 L/min is also used to determine the mass and chemical composition of fine particles (Du W. et al., 2017; Fan Z.L. et al., 2017; Thuy N.T.T. et al., 2018) or used as the reference for calibration (Begum B.A. and Hopke P.K., 2019). However, the previous studies showed that the MiniVol data did not agree well with the other collocated samplers and monitors (Kingham S. et al., 2006). This could be due to the particle bounce and particle loading effect. The impaction plate of the MiniVol impactor needs to be cleaned and recoated grease after 5 sampling days or more often if overloading is observed to recovery the good performance (AMI, 2020). The mini-volume (4 L/min) or low-volume (10–20 L/min) Harvard sharp-cut impactors with different cut-sizes of 10, 2.5 and 1.0 µm, which use the oil-coated impaction substrate to eliminate the particle bounce effect, were mainly used for indoor measurement and outdoor field tests (Rovelli S. et al., 2017).

3.1.2 The inlet for two size fractions

To determine the compliance with NAAQS, it is often desirable to fractionate particles into two size fractions simultaneously, one corresponding to the coarse particle mode (> 2.5 µm) and one corresponding to the fine particle mode (≤ 2.5 µm). The virtual impactor is very suitable for this application (Lin G.Y. et al., 2015; Landis M.S. et al., 2017). The Andersen Dichotomous Virtual impactor (McFarland A.R. et al., 1978) is the most widely used for collecting coarse and fine particles at a low flow rate of 16.7 L/min (Gugamsetty B. et al., 2012). Other types of virtual impactors were developed and deployed for specific measurement purposes. For instance, the $PM_{1.0/2.5/10}$ trichotomous sampler (cascade virtual impactor) with the high sampling flow rate of 1133 L/min was developed to measure the mass and the ion concentrations of PM at Phoenix, Arizona, USA (Marple V.A. et al., 2013, 2014a).

3.2 Particle mass distribution measurement

The determination of the mass concentration of the particles less than a certain size is valuable but it is desirable to determine the mass distribution of entire aerosol particles. In this case, the cascade impactor can be deployed to cover a very wide particle size range from a few nanometers to tens of micrometers (Marple V.A. et al., 1991; Fernandez de la Mora J. and Schmidt-Ott A., 1993; Arffman A. et al., 2015). It is desirable to have a cascade impactor for particle mass distribution measurement with a wide range of flow rates, a large range of particle sizes, a good size resolution (or number of channels), low pressure drop, uniform particle collection, less inter-stage loss, no particle bounce, high loaded mass, and easy assembly and disassembly of the stages (Marple V.A., 2004). The list of the commercially available cascade impactors is shown in **Table 8-1** in Marple V.A. and Olson B.A. (2011). Some commonly used cascade impactors are Andersen cascade impactor (Andersen A.A., 1966), Mercer cascade impactor (Mercer T.T. et al., 1970), Quartz Crystal Microbalance (QCM) cascade impactor (Hering S.V., 1987), Pilat cascade impactor (Pilat M.J. et al., 1970), Berner low-pressure cascade impactor (Berner A., 1972), Low-Pressure Impactor (Berner A., 1972), Dekati® Low-Pressure Impactor (DLPI) (Durand T. et al., 2014), MOUDI (Marple V.A. et al., 1991; Marple V.A. et al., 2014b) and NMCI (Liu C.N. et al., 2013b; Chien C.L. et al., 2015).

The most widely used cascade impactor for determining particle mass distribution is MOUDI, which measures mass concentrations in ten size fractions down to nano-size particles with a moderate pressure drop (D_{pa50} = 10, 5.6, 3.2, 1.8, 1.0, 0.56, 0.32, 0.18, 0.10, and 0.056 μm) at the medium flow rate of 30 L/min (Marple V.A. et al., 2014b). In the MOUDI, particles with different size ranges are collected on the well-prepared substrates such as oil-coated aluminum foil or Teflon filter to determine the mass concentration distribution. An advantage of the MOUDI is that it can collect particles uniformly on the substrate by using multiple nozzles and rotating impaction plates and/or nozzle plates. The impaction substrates can either be Teflon filters for mass determination and chemical composition or grease-coated aluminum foil for mass concentration only without significant particle bounce effect. The MOUDI coupled with nano-MOUDI or 10 L/min MOUDI-II was further developed to have higher resolution for nanoparticles with three additional stages (D_{pa50} = 0.010, 0.018 and 0.032 μm) after the 10th stage (D_{pa50} = 0.56 μm) (Marple V.A. and Olson B.A., 1999). The MOUDI-II is used for size-fractionated measurement of mass concentration, chemical composition, and morphology of ambient ultrafine particles. The MOUDI-II needs a long sampling time to collect enough NP mass to measure their mass concentrations and chemical compositions accurately. Our group studied the performance of the MOUDI and found that the nozzles of 7th–10th stages of the MOUDI are clogged easily due to the stepwise structure with abrupt contractions. The nozzle plates are not suitable for ultrasonic cleaning to remove particle deposits due to their thin and fragile structure (Liu C.N. et al., 2013b). Therefore, the NMCI with a smooth nozzle shape was developed to eliminate the clogging problems (Liu C.N. et al., 2013b).

Due to the need for the automatic measurement of particle mass distribution, several cascade impactors have been developed and commercialized. The requirement of the real-time measurement is the sensitivity for low particle concentration detection and short response time. The ten-stage QCM cascade impactor, in which piezo-electric quartz crystals sensors are flush-mounted on the impaction plates, was developed earlier (Chuan R.L., 1970). In the QCM cascade impactor, the mass concentrations are calculated based on the difference in the frequencies of the QCMs before and after particle impaction on the piezoelectric crystal (Chen M. et al., 2016). The previous studies showed that the QCM cascade impactor has the particle bounce effect (Hering S.V., 1987), high particle loss of up to 50 %, low off-center sensitivity, and low sensitivity of nanoparticle concentrations (Keskinen J. et al., 1992). A novel QCM cascade impactor was developed to overcome these problems, in which the RH controller was used to control the RH of incoming aerosol to between 40 % and 65 % to eliminate the particle bounce effect and ensure particle coupling with the QCM, and the impedance measurement was used to achieve nanogram mass resolution (Chen M. et al., 2016).

The Electrical Low-Pressure Impactor (ELPI) developed by Keskinen J. et al. (1992) can be used to detect the number concentration distribution in real-time. The ELPI consists of a unipolar charger followed by a cascade impactor coupled with multichannel electrometers. Particles are first charged unipolarly when they pass through the charger and then collected on the electrically isolated impaction plates in the cascade impactor. The current of each stage is read by a corresponding electrometer channel and converted to particle number concentration. Numerous studies have been conducted to calibrate and improve the performance of the ELPI. The latest ELPI (Dekati® ELPI®+) was developed to measure particle number distribution with very high size resolution (13 channels), wide particle size range (from 6 nm to 10 μm), and fast response (seconds) (Järvinen A. et al., 2014). It should be noticed that the performance of the ELPI can be affected by the charging efficiency which depends on particle size and concentration and is influenced by the aging of the charger (Järvinen A. et al., 2014). The electrical low-pressure cascade impactor was developed further for achieving higher resolution with 100 or 500 size fractions (i.e. HR-ELPI+) (Saari S. et al., 2018) or measuring nanoparticle mass concentration using a wire-to-rode corona charger to eliminate the new particle formation which may occur in the pin-to-plate corona charger (Han J. et al., 2018).

3.3 Personal exposure measurement

Personal exposure measurement of PM is to assess the PM exposure of individual workers in workplaces or microenvironments where they spend most of their time daily (Marple V.A. and Olson B.A., 2011). The personal exposure measurement depends on the pollutant concentration and exposure time. Comparison to the inertial impactors used for PM sampling, the inertial impactors which are used for personal exposure measurement have specific requirements. The devices have to be light, small, portable, self-contained and battery-powered to move easily from one location to another and are placed near the personal breathing zone, which is about 30 cm hemisphere around mouth and nose (Asbach C. et al., 2017). These devices need to collect an adequate amount of particle mass with a uniform deposit on the substrate for subsequent PM mass concentration, chemical composition, morphological, and crystallographic analyses manually. The personal exposure devices are listed in Volkwein J.C. et al. (2011) for workplaces and in Rodes C.E. (2011) for non-occupational microenvironments. Personal exposure impactors measure PM_{10}, $PM_{2.5}$, and $PM_{1.0}$, inhalable (D_{pa50} = 100 µm, thoracic (D_{pa50} = 10 µm), and respirable (D_{pa50} = 4 µm) particles regulated by the National Institute for Occupational Safety and Health (NIOSH), International Organization for Standardization (ISO), and Comite' European de Normalisation (CEN) (Volkwein J.C. et al., 2011).

To meet the respirable aerosol sampling criteria, the inertial impactors are designed to have a less sharp efficiency curve (Volkwein J.C. et al., 2011). Personal single-stage impactor (Tsai C.J. et al., 2012b), two-stage impactor (Demokritou P. et al., 2001), three-stage personal impactor (Tsai C.J. et al., 2008), and cascade impactor (Chen M. et al., 2018) and multipollutant samplers were developed for personal exposure measurement since 1970s and some which were widely used. These include Marple personal impactor, personal environmental monitor (PEM), mini-MOUDI, parallel particle impactor, personal modular impactor, Sioutas cascade impactor, personal PUF pesticide sampler, and personal impactor filter pack (Marple V.A. and Olson B.A., 2011). The Marple personal impactor with 8 stages at the flow rate of 2 L/min and Sioutas cascade impactor at the flow rate of 9 L/min are the most frequently used samplers for mass distribution measurement (Tsai C.J. et al., 1997b; Sharma R. and Balasubramanian R., 2018).

The personal exposure measurement for nanoparticles has drawn serious concern due to the adverse health effect of nanoparticles released from nanopowders and nanomaterial manufacturing processes. Nanopowders such as TiO_2, SiO_2, and ZnO are released to the workplaces depending on their dustiness or the property of a given material to generate dust (Tsai C.J. et al., 2009; Tsai C.J. et al., 2011). Some studies were carried out by our group (Tsai C.J. et al., 2011; Tsai C.J. et al., 2012a) to determine the generation and dispersion characteristics of nanoparticles.

Some commercialized personal cascade impactors can be used for the nanoparticle exposure measurement such as mini-MOUDI (13 stages with particle sizes ranging from 10 nm to 10 µm) operating at the flow rate of 2 L/min and Sioutas cascade impactor (4 stages with particle sizes ranging from 0.25 to 2.5 µm) operating at the flow rate of 9 L/min (Asbach C. et al., 2017). Due to the high pressure drop that is needed to classify nanoparticles, only a few of personal exposure samplers were developed to collect nanoparticles. Besides, the nanoparticle samplers need to operate for a long sampling period to collect enough mass for further gravimetric analysis since nanoparticle mass is very small. The personal cascade impactors using inertial filters for nanoparticle collection were developed with a moderate pressure drop (< 5–10 kPa) (Kumsanlas N. et al., 2019) which is based on the design of inertial filter (Zhang T. et al., 2017).

Our group developed the PENS, which can be used to collect respirable particles and nanoparticles simultaneously by using a cyclone as a pre-separator to remove particles larger than 4 µm and a micro-orifice impactor with the D_{pa50} of 101.4 nm followed by the filter cassette (Tsai C.J. et al., 2012b). The PENS eliminated the particle bounce and overloading effects by using a rotating, silicone oil-coated Teflon filter as the substrate. The PENS operates at a low flow rate of 2 L/min and slightly high pressure drop (< 14–15 kPa), which is still acceptable for a personal sampling pump of 21 kPa (AirChek XR5000, USA). The PENS are used for NP mass concertation measurement in workplaces (Young L.H. et al., 2013; Asbach C. et al., 2017) and ambient air as well (Zhu C.S. et al., 2016; Zhu C.S. et al., 2017). The field comparison test of the PENS with the collocated SKC respirable dust aluminum cyclone (SKC Inc., PA, USA) and sequential mobility particle sizer (SMPS; GRIMM Aerosol Technik, Gmbh, Germany, Model 5.500) in metal working places showed that the PENS correlated well with the cyclone for PM_4 and SMPS for $PN_{0.1}$ (Young L.H. et al., 2013). The aspiration efficiency of the PENS was good and agreed well with that of NIOSH two-stage personal bioaerosol samplers (Zhou Y. et al., 2014).

The PENS is currently being studied further to enhance its performance at a higher flow rate of 3–5 L/min and a lower pressure drop of < 15 kPa. To achieve the same cutsize of lower than 100 nm at the higher flow rate and lower pressure drop, the PENS can be re-designed with more nozzles with larger diameter and smaller S/W. The wetted GFF substrate can be used in the PENS to wash off deposited particles continuously to achieve a long sampling period without particle bounce and overloading effects (Le T.C. and Tsai C.J., 2017).

3.4 PM control

Control of PM emission is critical to reduce airborne PM concentration and health risks. Major particle control devices include wet scrubber, electrostatic precipitators (ESPs), cyclones, and baghouses, each of which has the advantages and disadvantages in the application. The wet scrubbers remove particles by using water spray or water film, which causes a very high operating pressure drop, metal corrosion, and water pollution (Huang C.H. et al., 2007; Pui D.Y.H. et al., 2014). The ESPs which use electrostatic force to remove particles have the minimum removal efficiency for particles ranging from 0.1 to 1.0 μm due to the minimum electrical mobility in this submicrometer range (Hinds W.C., 1999; Pui D.Y.H. et al., 2014). The cyclones are used extensively for PM control due to its simple structure, low pressure drop, and low cost but the collection efficiency for fine particles is low (Pui D.Y.H. et al., 2014). The baghouses use fabric filters to remove particles based on dust cake filtration with a high pressure drop (Hinds W.C., 1999; Pui D.Y.H. et al., 2014). Thus, the baghouses are required to be cleaned and regenerated frequently. Special materials such as electret filters can be used to enhance the efficiency of the depth filtration with a low pressure drop (Tien C.Y. et al., 2020).

Our group developed successfully some control devices overcoming some drawbacks to achieve high removal efficiency for fine and ultrafine particles including a wire-on-plate ESP (Li Z. et al., 2015), an efficient single-stage wet ESP (Lin G.Y. et al., 2010), an efficient venturi scrubber system (Tsai C.J. et al., 2005; Huang C.H. et al., 2007), and a wet electrocyclone (Lin G.Y. et al., 2013). The efficient venturi scrubber system uses a condensation device to grow sub-micron particles to super-micron sizes before they are removed by a traditional venturi scrubber operating at a moderate pressure drop of 4.3 kPa (Huang C.H. et al., 2007). The wet electrocyclone combining the cyclone with an electrostatic precipitator with wall cleaning by water film was developed to enhance the collection efficiency for ultrafine particles for a long service period (Lin G.Y. et al., 2013). This combination technique was referred by other researchers (Titov A., 2015; Zhang X. et al., 2018).

The well-designed control devices can be used for PM and NP control in industries but they are not suitable for some circumstances such as the semiconductor industry with the coexisting residual gases and ultrafine particles (Tsai C.J. et al., 1997a). For instance, the ESPs may cause fire hazard or explosion. The inertial impactor is a potential device to replace the conventional control devices for removing NPs with a smaller footprint than the wet scrubber, a higher efficiency than the cyclone, less costly than the ESP, and a longer service time than the baghouse since it can be designed to have ultrafine cut-sizes and can use water/oil-wetted impaction substrate to eliminate overloading effect without frequent maintenance. However, the impactor requires a high operating pressure drop for NP control. This can be resolved by growing particles by condensation method. Recently, by coupling with particle condensation growth method, the inertial impactors were deployed to control ultrafine particles as a filterless control method to save the cost (Pyo J. et al., 2017). This condensation technique was used in the past to enlarge ultrafine particles to submicron particles to increase the collection efficiency of the impactor (Demokritou P. et al., 2002a). The combination of the inertial impactor with the particle growth device has the potential for ultrafine particle removal with a high removal efficiency, small footprint, moderate pressure drop and long-term use without frequent maintenance need.

3.5 Powder classification by inertial devices

Powder classification is different from PM sampling since powder sources generate high mass concentration and wide size range (0.01 to 1000 μm) (Lai W.H. et al., 2005). Powder classification is to separate powder smaller than a certain size from the source powder to obtain desired narrow powder distributions with high precision to use in bio-medicine, electronic information, fine ceramics, food industry, and coating industry (Guo L. et al., 2007; Wang D. et al., 2014). Powder classifiers can also be designed to classify coarse, fine, and nano-sized particles simultaneously with precise cutoff sizes and sharp cutoff characteristics to achieve high production rates (Liang S., 2010). Classification by inertial/centrifugal force such as cyclone is one of the popular methods since the cyclone has a simple structure and cheaper to make (Yoshida H. et al., 1991; Yoshida H. et al., 2008). It is found that large particles are repulsed or re-entrained if the inner wall of the cyclone is not oil-coated (Yoshida H. et al., 1991), and particles smaller than cut-size will be collected due to the turbulent eddies in hydrocyclone resulting in the so-called fishhook effect of the powder classification efficiency curve (i.e., the raise of curve in small particle size range) (Roldanvillasana E. et al., 1993; Lai W.H. et al., 2005). It is to be noted that the sharpness mentioned here refers to the ratio of D_{pa75}/D_{pa25} (Morimoto H. and Shakouchi T., 2003; Lai W.H. et al., 2005).

Nowadays, powder classifiers are developed to separate nanopowder with a sharp classification efficiency (i.e., without contamination of coarse powders and fishhook effect) to have a uniform powder to use in ultra-conductor, optical, thermal, electromagnetic, and biomedical fields (Morimoto H. and Shakouchi T., 2003; Wang D. et al., 2014). In comparison to the cyclones, the inertial impactors, which have sharp cutoff characteristics and accurate cut-size, have a good potential to be used for powder classification with a narrow size range. The water/oil wetted inertial impactor in which the impaction substrate is washed clean by using water or oil can be used for powder classification at a high particle loading rate without particle loading effect. Further investigation to use the impactors for power classification is worthwhile.

4. Conclusions and future perspective

The study of the inertial devices has been an interesting topic since the 1860s when the first impactor was developed to observe the relationship between the particles and health effects due to its simple principle and structure, easy operation, and low cost. The design theory was also developed numerically based on the critical experimental and field tests by Marple and his co-workers. As a result, many inertial impactors were commercialized to serve as the size-fractionators and samplers for mass concentration measurement, mass and number distribution measurement, particle collection for chemical analysis, and health exposure measurement. Numerous studies were conducted to resolve the particle bounce, particle loading, and particle loss effects to improve the performance of the inertial impactors. Therefore, the inertial impactors are deployed widely to measure a wide size range of particles with a wide range of concentration and different particle types.

Inertial impaction technique still attracts researchers to work on it since the application of inertial devices is diverse. The impactors can be studied further to extend the cut-size to a very small size such as nano-size, provide real-time mass and number concentrations with very high sensitivity, fast response, and high resolution, and apply for specific applications in medical and pharmaceutical areas (Marple V.A., 2004). Besides sampling and measurement, the inertial impactors can potentially be used for PM control with a lower cut-size than the cyclone, a lower pressure drop than the baghouse and less costly than the electrostatic precipitator if they are coupled with condensation particle growth devices. The biggest challenge of the inertial impactors for the control and powder classifications is how to scale up and extend the lifetime. To meet these challenges, this article has reviewed and provided the design considerations and applications of the inertial impactors for the benefit of aerosol and powder communities.

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Nomenclature

C_c	Slip correction factor
D	Collection probe diameter (m)
D_p	Particle diameter (m)
D_{pa50}	Cutoff aerodynamic diameter (m)
GSD	Geometric standard deviation
μ	Dynamic air viscosity (kg·s/m^2)
ρ_a	Air density (kg/m^3)
ρ_p	Particle density (kg/m^3)
ρ_{p0}	Unit density (1000 kg/m^3)
Q	Total flow rate (L/min)
Q_o	Major flow rate (L/min)
Q_1	Minor flow rate (L/min)
Re	Reynolds number
S	Jet-to-plate distance (m)
Stk	Stokes number
Stk_{50}	Stk for 50 % collection efficiency
T	Throat length (m)
U_0	Air jet velocity (m/s)
W	Nozzle diameter/width (m)

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